QP-Unit2

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The Born interpretation

- **What is a wave function?**
- It has all the dynamical information about the particle.
- More immediately, it has the information about the location of the particle.



Max Born

The Born interpretation

- The square of the wave function $|\Psi|^2$ at a point is proportional to the probability of finding the particle at that point.

$$|\Psi|^2 = \Psi^* \Psi \geq 0$$

Always real, non-negative

Complex conjugate of Ψ

$$|a+bi|^2 = (a+bi)^*(a+bi) = (a-bi)(a+bi)$$
$$= a^2 - b^2 i^2 = a^2 + b^2 \geq 0$$

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The Born interpretation

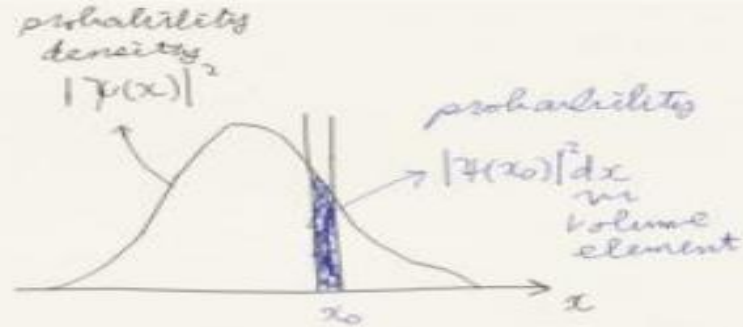
- A wave function is in general complex.
- But $|\Psi|^2$ is always real, non-negative.



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The Born interpretation

- One-dimension: if the wave function of a particle has the value Ψ at point x, the probability of finding the particle between x and $x+dx$ is proportional to $|\Psi|^2\, dx$.



Normalization

- We are free to multiply any constant factor (other than zero) to Ψ, without stopping it from the solution of the Schrödinger equation.
- Remembering that $|\Psi|^2 dxdydz$ is only **proportional** to the probability of finding the particle in $dxdydz$ volume at (x,y,z), we consider it the most desirable and convenient if the wave function be **normalized** such that finding the particle somewhere in the space is equal to 1.

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Normalization

- We multiply a constant to Ψ.

$$\Psi' = N\Psi$$

such that

$$\int |\Psi'|^2 dx = N^2 \int |\Psi|^2 dx = 1 \qquad \iiint |\Psi'|^2 dxdydz = N^2 \iiint |\Psi|^2 dxdydz = 1$$

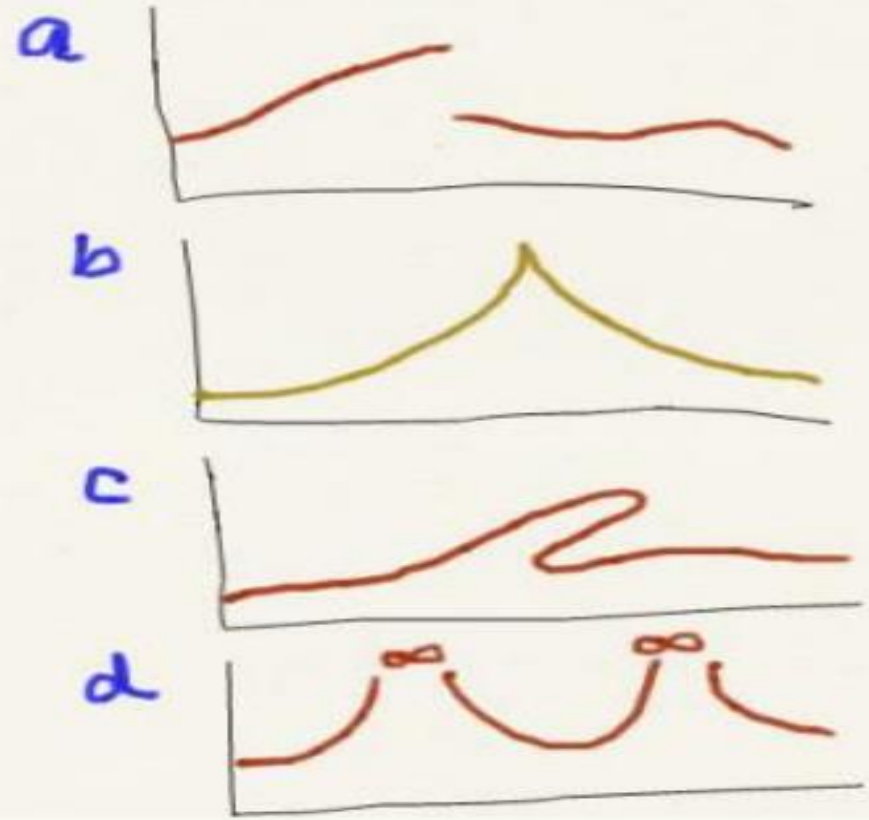
- These equations mean that probability of finding the particle somewhere is 1. After normalization, $|\Psi|^2 dxdydz$ is not only proportional but is **equal** to the probability of finding the particle in the volume element $dxdydz$ at (x,y,z).

- If Ψ is a normalized solution of time-independent SE, Ψe^{ik} for any real value of k is also a normalized solution of SE because

$$\left|\Psi e^{ik}\right|^2 = \left(\Psi e^{ik}\right)^* \left(\Psi e^{ik}\right) = \left(\Psi^* e^{-ik}\right)\left(\Psi e^{ik}\right)$$
$$= \left|\Psi\right|^2 e^{-ik} e^{ik} = \left|\Psi\right|^2$$

- The simplest example is when $e^{i\pi} = -1$. Ψ and $-\Psi$ are both normalized and with the same probability density $|\Psi|^2$.

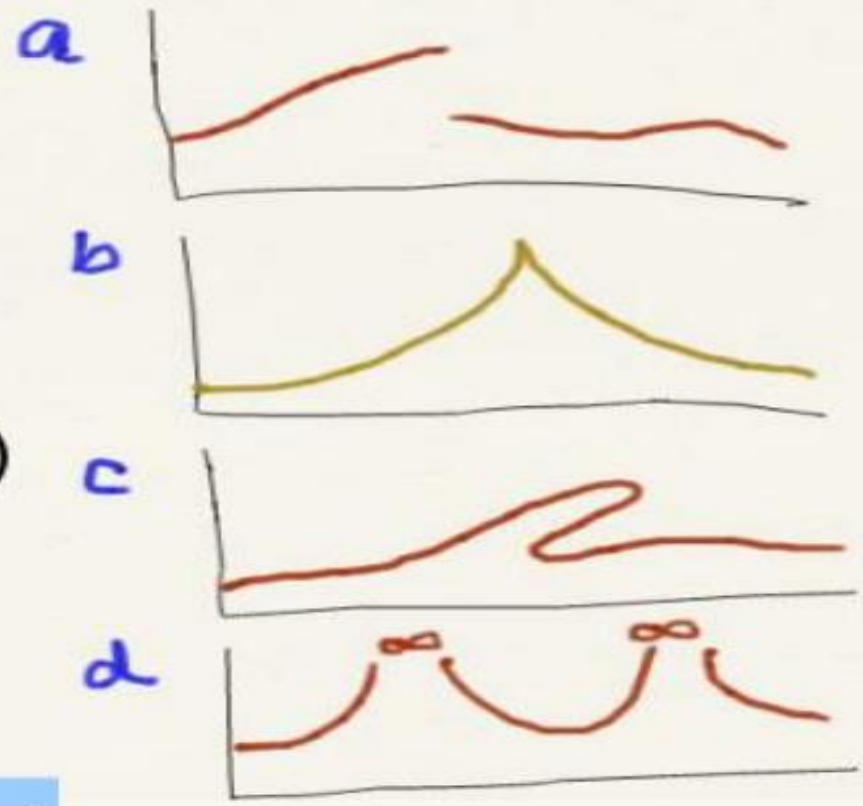
- **The Born interpretation:** the square of a wave function is a probability density.
- This immediately bars a wave function like figure (c), because a probability should be a unique value (**single valued**)



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- Probability should add up to unity, when all possibilities are included. Square of a wave function should integrate to unity.
- This bars a function like (d) because it integrates to infinity regardless of any nonzero normalization constant (**square integrable**).

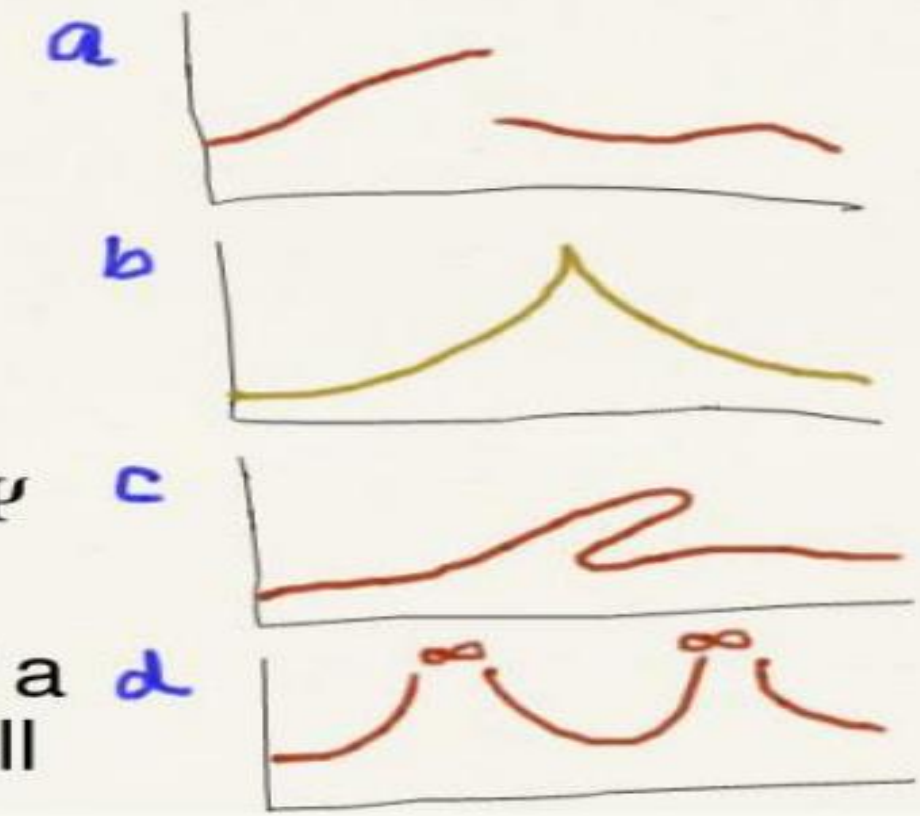
$$\int |\Psi'|^2 \, dx = 1$$



- Apart from the Born interpretation, the form of the Schrödinger equation itself set some conditions for a wave function.

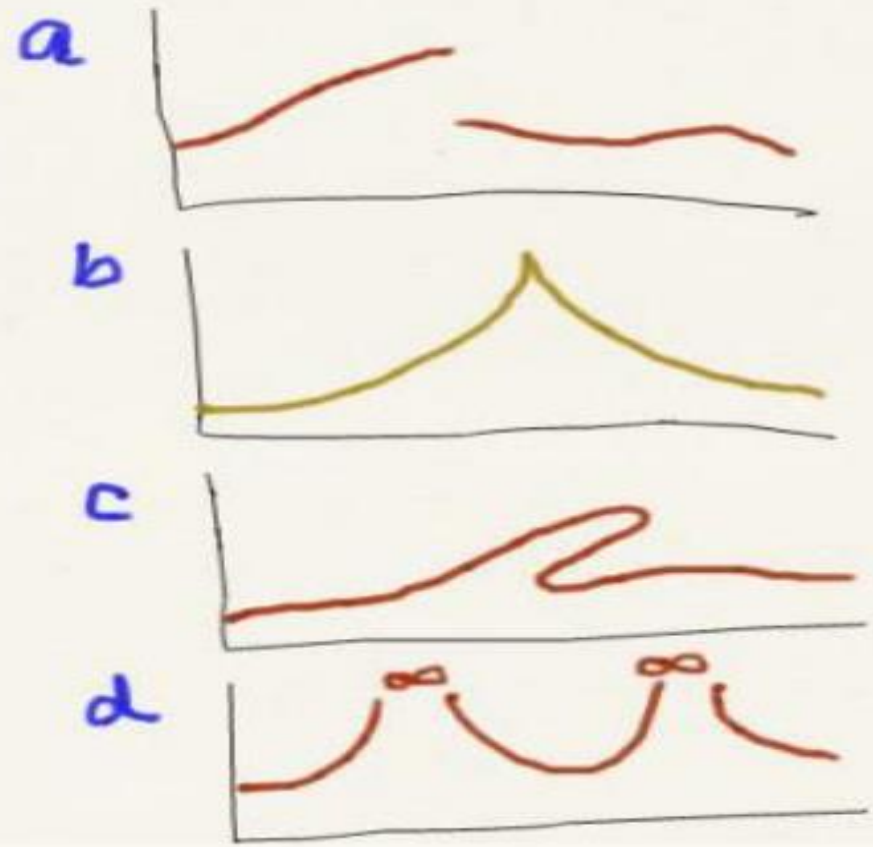
$$\hat{H}\Psi = \left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \hat{V}(x) \right]\Psi = E\Psi$$

- The second derivatives of a wave function must be well defined.

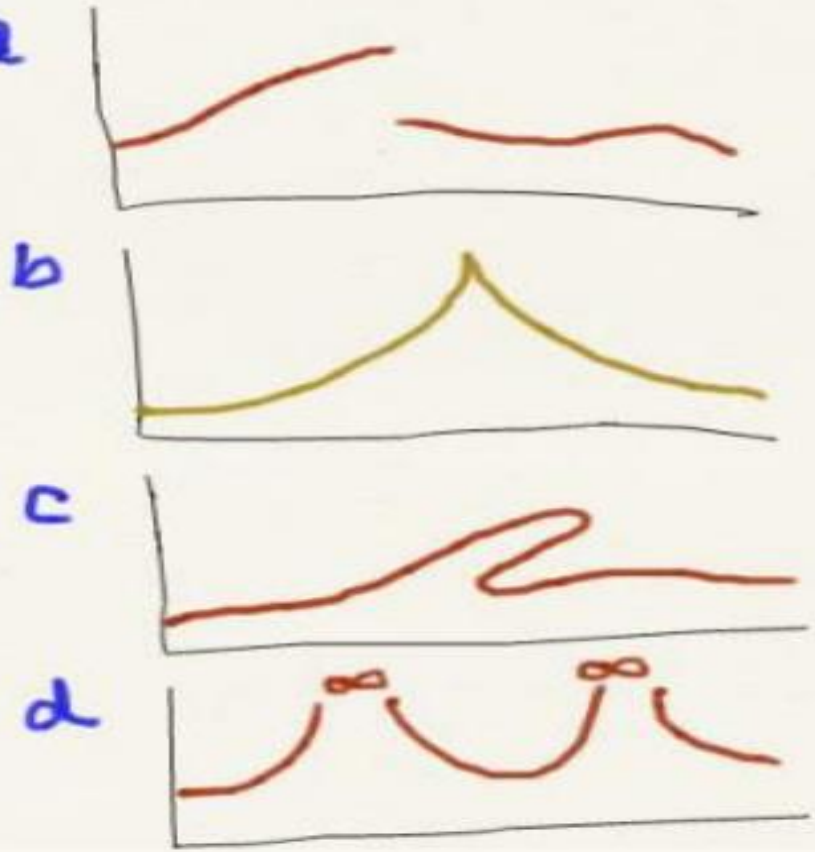


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- For the second derivative to exist, the wave function must be **continuous**, prohibiting a function like (a) which is discontinous.
- It is also impossible to imagine a system where the probability density changes abruptly.



- For the second derivatives to be nonsingular, the wave function should usually be **smooth**, discouraging a kinked function like (b).
- There are exceptions. When the potential *V* also has a singularity, a kinked wave function is possible.



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